

Project target

1 Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset-

(1) Data type of all columns in the "customers" table.

Code-

```
select column_name,
data_type
from `target_ecom.INFORMATION_SCHEMA.COLUMNS`
where TABLE_NAME='customers'
```

output

column_name	data_type
customer_id	STRING
customer_unique_id	STRING
customer_zip_code_prefix	INT64
customer_city	STRING
customer_state	STRING

(2) Get the time range between which the orders were placed.

Code –

```
select
min(order_purchase_timestamp)as `First_order_placed`,
max(order_purchase_timestamp)as `Last_order_placed`
from `target_ecom.orders`
```

output

First_order_placed	Last_order_placed
2016-09-04 21:15:19.000000 UTC	2018-10-17 17:30:18.000000 UTC

TARGET SALES 2017-2018 ANALYSYS

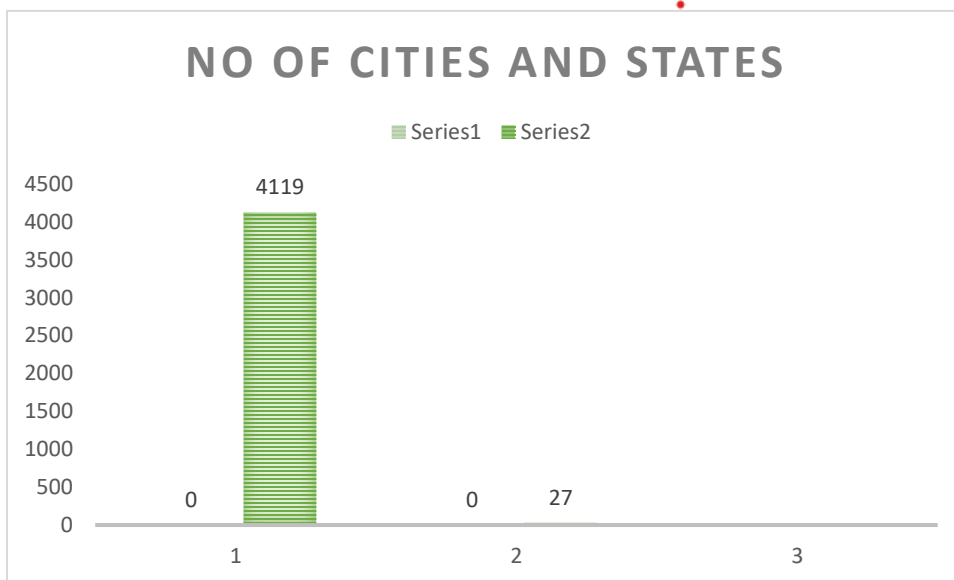
(3) Count the Cities & States of customers who ordered during the given period.

Code –

```
select  
count(distinct customer_city)as`no_of_cities`,  
count(distinct customer_state )as `no_of_states`  
from `target_ecom.orders` o inner join `target_ecom.customers` c  
on o.customer_id=c.customer_id
```

output

1	no_of_cities	no_of_states
2	4119	27
3		
4		
5		



2. In-depth Exploration

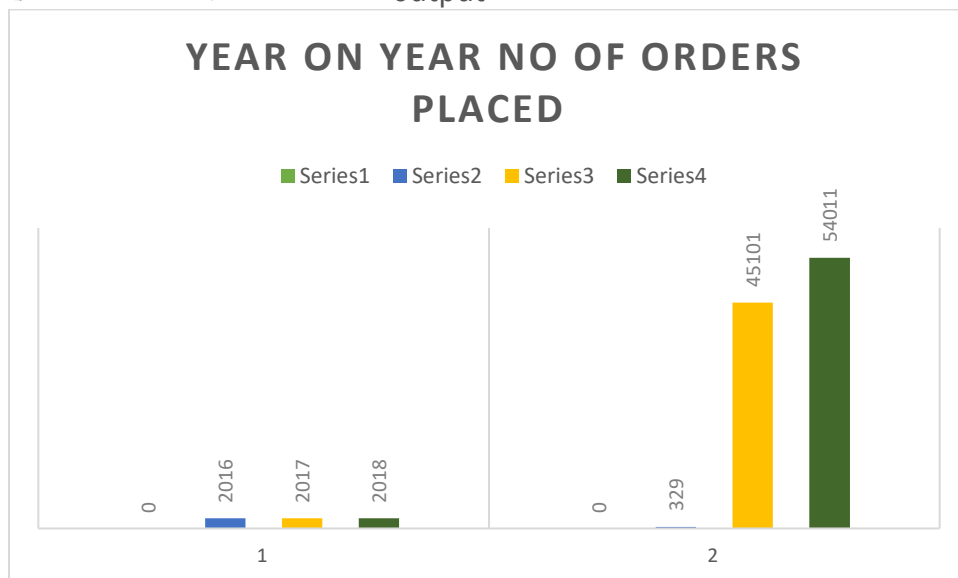
(1) Is there a growing trend in the no. of orders placed over the past years?

TARGET SALES 2017-2018 ANALYSYS

```
select
extract(year from order_purchase_timestamp)as `year`,
count(order_id) as `no_of_orders`
from `target_ecom.orders`
group by
year
order by
year;
```

year	no_of_orders
2016	329
2017	45101
2018	54011

output



2.(2) Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Code

```
select
extract(month FROM order_purchase_timestamp) as `month`,
count(order_id)as`no_of_orders_placed`
from
`target_ecom.orders`
group by
month
order by
no_of_orders_placed desc;
```

TARGET SALES 2017-2018 ANALYSYS

output

1	month	no_of_orders_placed
2	1	8069
3	2	8508
4	3	9893
5	4	9343
6	5	10573
7	6	9412
8	7	10318
9	8	10843
10	9	4305
11	10	4959
12	11	7544
13	12	5674
14		

Chart



Findings

1. There is an increase in sales during winter season and summer season.
2. sales decrease in month of September and October.
3. maximum orders placed in the month of may and august.

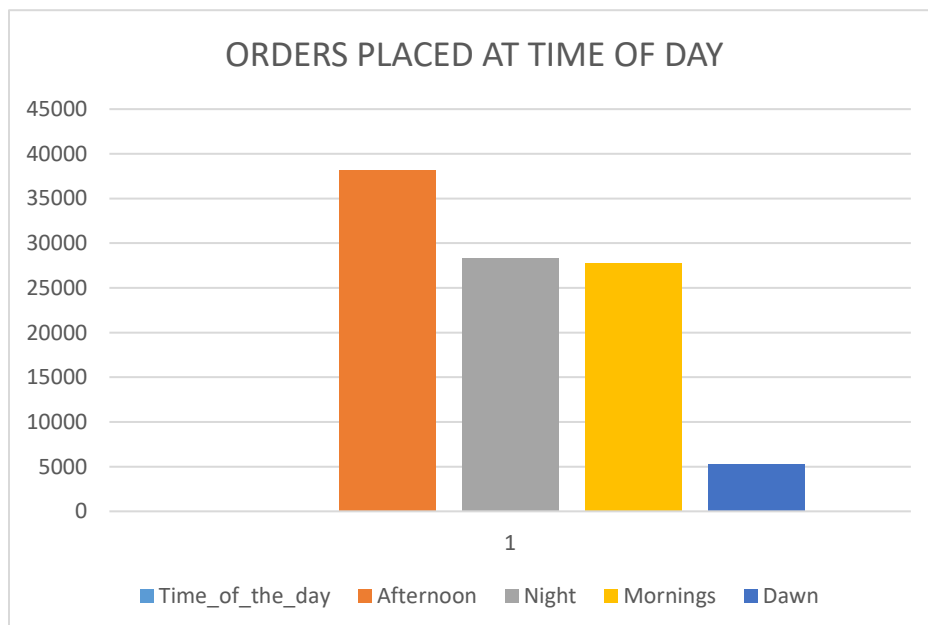
TARGET SALES 2017-2018 ANALYSYS

2 (3) During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

code

```
select
( case when extract(hour from order_purchase_timestamp ) between 0 and 6 then 'Dawn'
      when extract(hour from order_purchase_timestamp ) between 7 and 12
then'Mornings'
      when extract(hour from order_purchase_timestamp ) between 13 and 18
then'Afternoon'
      when extract(hour from order_purchase_timestamp ) between 19 and 24 then 'Night'
end
)as `Time_of_the_day`,
count(order_id)as`no_of_orders_as_time`
from
`target_ecom.orders`
group by
Time_of_the_day
order by
no_of_orders_as_time desc;
```



TARGET SALES 2017-2018 ANALYSYS

1	Time_of_the_day	no_of_orders_as_time
2	Afternoon	38135
3	Night	28331
4	Mornings	27733
5	Dawn	5242
6		
7		

Findings

- 1 During Afternoon time of the day, Brazilian customers mostly place their orders.
2. 1 During dawn time of the day, Brazilian customer place their orders minimum.

3 Evolution of E-commerce orders in the Brazil region:

- (1) Get the month on month no. of orders placed in each state.

Assumption

We should extract month from order table.

#then count no of orders on order id.

group by code on state and month.

#inner join order and customer table to get no of orders and state.

Code

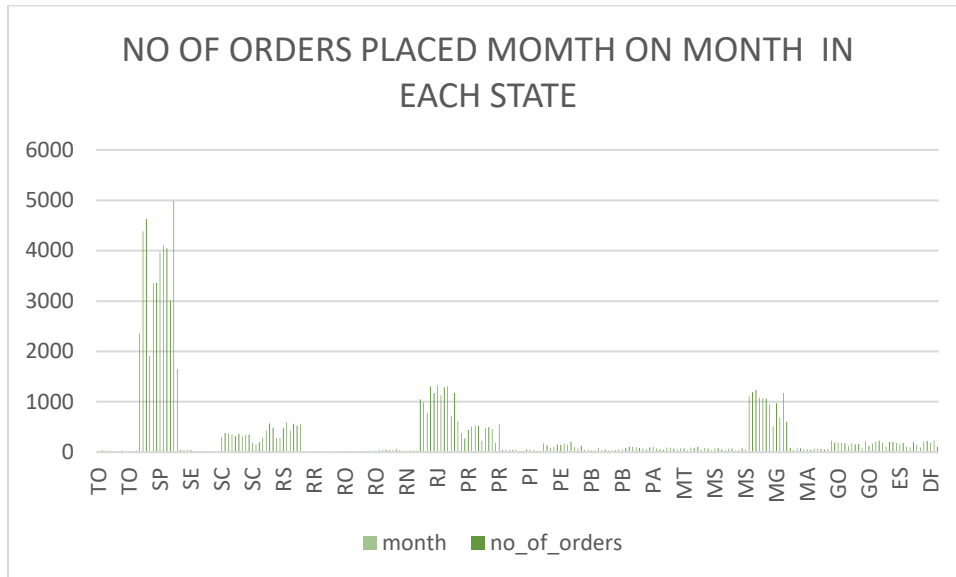
```
select
c.customer_state,
extract(month from o.order_purchase_timestamp ) as `month`,
count(o.order_id) as `no_of_orders`
from
`target_ecom.orders` o inner join `target_ecom.customers` c
on o.customer_id =c.customer_id
group by
c.customer_state,month;
```

output table

TARGET SALES 2017-2018 ANALYSYS

1	customer_	month	no_of_orders	
2	SP	8	4982	
3	SP	5	4632	
4	SP	7	4381	
5	SP	6	4104	
6	SP	3	4047	
7	SP	4	3967	
8	SP	2	3357	
9	SP	1	3351	
10	SP	11	3012	
11	SP	12	2357	
12	SP	10	1908	
13	SP	9	1648	
14	RJ	5	1321	
15	RJ	8	1307	
16	RJ	3	1302	
17	RJ	7	1288	
18	MG	3	1237	
19	MG	5	1190	
20	MG	8	1177	
21	RJ	2	1176	
22	RJ	4	1172	
23	RJ	6	1128	
24	MG	7	1111	
25	MG	6	1080	
26	MG	2	1063	
27	MG	4	1061	
28	RJ	11	1048	
29	RJ	1	990	
30	MG	1	971	
31	MG	11	812	

TARGET SALES 2017-2018 ANALYSYS



Findings-

--- In this month on month analysis we can see state SP placed maximum orders.

no of orders in state SP higher in July, august and April month

minimum order placed by states AC and AP

Suggestions

target need to focus on states where min order were placed.

3(2) How are the customers distributed across all the states.

Assumptions

we should count no of customers.

#group them by states.

Code

```
select
customer_state,
count(customer_id)as `no_of_customers`
from
`target_ecom.customers`
group by
customer_state
```

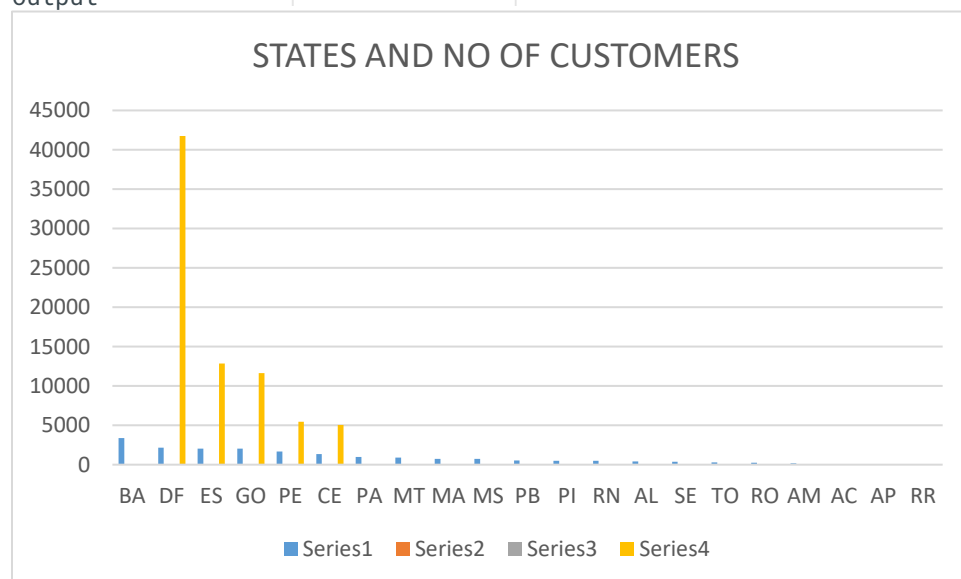

TARGET SALES 2017-2018 ANALYSYS

```
order by  
no_of_customers desc;
```

TARGET SALES 2017-2018 ANALYSYS

customer_state	no_of_customers
SP	41746
RJ	12852
MG	11635
RS	5466
PR	5045
SC	3637
BA	3380
DF	2140
ES	2033
GO	2020
PE	1652
CE	1336
PA	975
MT	907
MA	747
MS	715
PB	536
PI	495
RN	485
AL	413
SE	350
TO	280
RO	253
AM	148
AC	81
AP	68
RR	46

output



TARGET SALES 2017-2018 ANALYSIS

findings –

no of customers are more from SP RJ MG RS .

no of customers min at AC ,AP ,RR STATES.

4 Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

- (1) Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

Assumptions

to get year and month from use orders table.

to get total payment use payment table and payment_values column.

to get total payment in 2017 and 2018 between January to august use filter.

#To get percentage increase use lead/lag function and group it with order_year.

```
Code with `yearly_payments` as (select
    extract(year from o.order_purchase_timestamp ) as `order_year`,
    sum(p.payment_value) as `total_amount`
    from
        `target_ecom.orders` o inner join `target_ecom.payments` p on
o.order_id=p.order_id
    where
        (extract(year from o.order_purchase_timestamp ) between 2017 and 2018 )
        and
        (extract(month from o.order_purchase_timestamp ) between 1 and 8 )

    group by order_year
    order by order_year),

`increase_in_payments` as(select
                                order_year,
                                total_amount, lag(total_amount) over(order by
order_year asc) as `previous_yr_amt`
                                from yearly_payments)

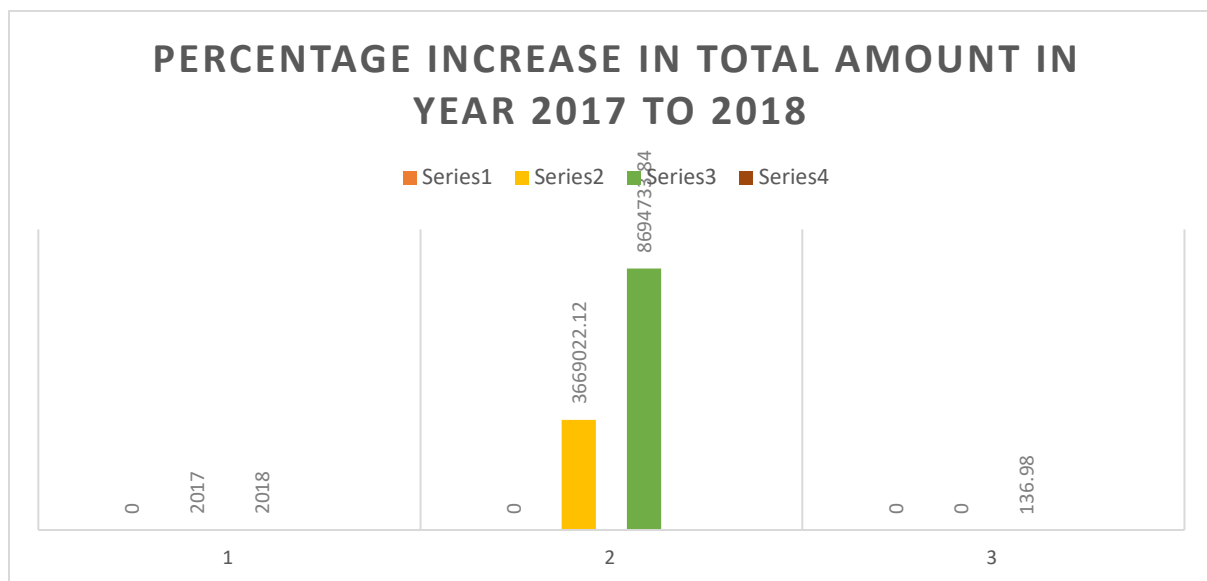
    select
        order_year, total_amount,
        round((((total_amount-
previous_yr_amt)/previous_yr_amt)*100),2) as `percentage_increase`
    FROM increase_in_payments order by order_year asc;
```

TARGET SALES 2017-2018 ANALYSYS

Output

1	order_year	total_amount	percentage_increase
2	2017	3669022.12	null
3	2018	8694733.84	136.98
4			
5			

CHART



findings # year 2017 to 2018 sales increase by 136%

4(2) Calculate the Total & Average value of order price for each state.

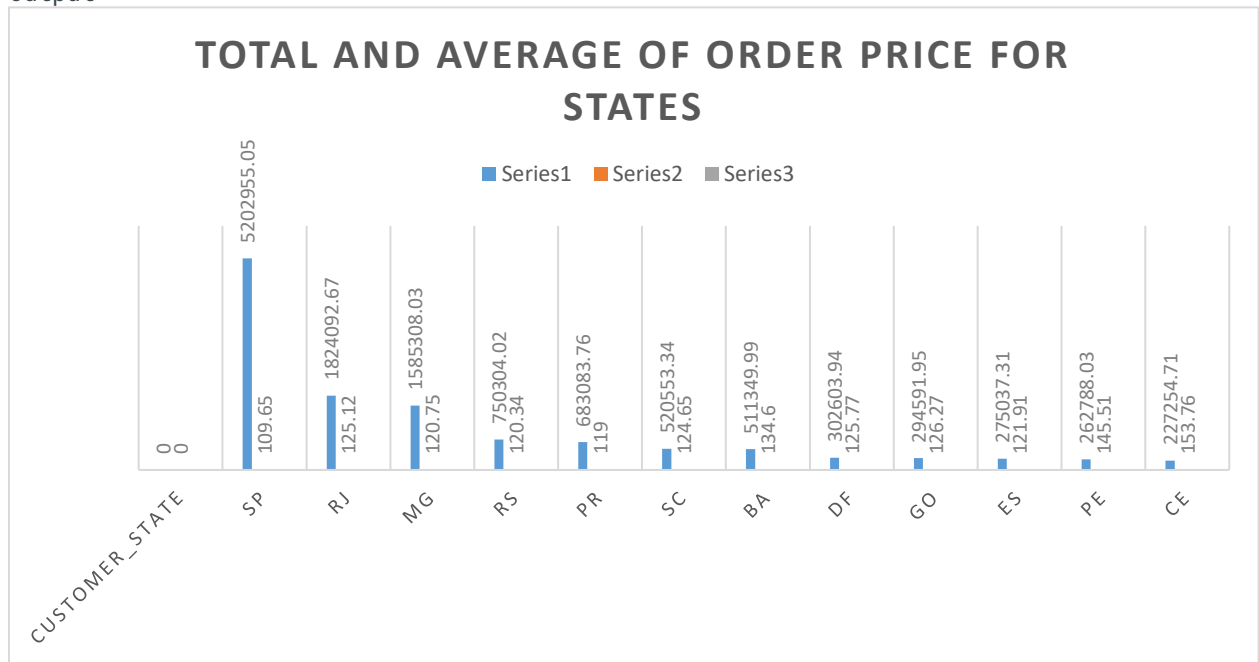
Code

```
#B. Calculate the Total & Average value of order price for each state.
select
c.customer_state,
round(sum(ot.price),2) as `total_value_of_order`,
round(avg(ot.price),2) as `avg_value-of_order`
```

TARGET SALES 2017-2018 ANALYSYS

```
from
`target_ecom.order_items` ot inner join `target_ecom.orders` o
on ot.order_id= o.order_id
inner join `target_ecom.customers` c on o.customer_id=c.customer_id
group by
c.customer_state
order by
total_value_of_order desc;
```

output



TARGET SALES 2017-2018 ANALYSYS

1	customer_	total_value_of_ord	avg_value-of_order
2	SP	5202955.05	109.65
3	RJ	1824092.67	125.12
4	MG	1585308.03	120.75
5	RS	750304.02	120.34
6	PR	683083.76	119
7	SC	520553.34	124.65
8	BA	511349.99	134.6
9	DF	302603.94	125.77
10	GO	294591.95	126.27
11	ES	275037.31	121.91
12	PE	262788.03	145.51
13	CE	227254.71	153.76
14	PA	178947.81	165.69
15	MT	156453.53	148.3
16	MA	119648.22	145.2
17	MS	116812.64	142.63
18	PB	115268.08	191.48
19	PI	86914.08	160.36
20	RN	83034.98	156.97
21	AL	80314.81	180.89
22	SE	58920.85	153.04
23	TO	49621.74	157.53
24	RO	46140.64	165.97
25	AM	22356.84	135.5
26	AC	15982.95	173.73
27	AP	13474.3	164.32
28	RR	7829.43	150.57
29			

findings

state SP ,RJ, AP shows maximum value of total order price .

state TO,RO,AM,AC,AP,RR shows the minimum total price of sales.

#PB state shows maximum avg order value.

SP state shows maximum price of total order but minimum of avg value of order price.

4(3) Calculate the Total & Average value of order freight for each state.

TARGET SALES 2017-2018 ANALYSYS

APPROACH-

sum of fright value.

avg of fright value .

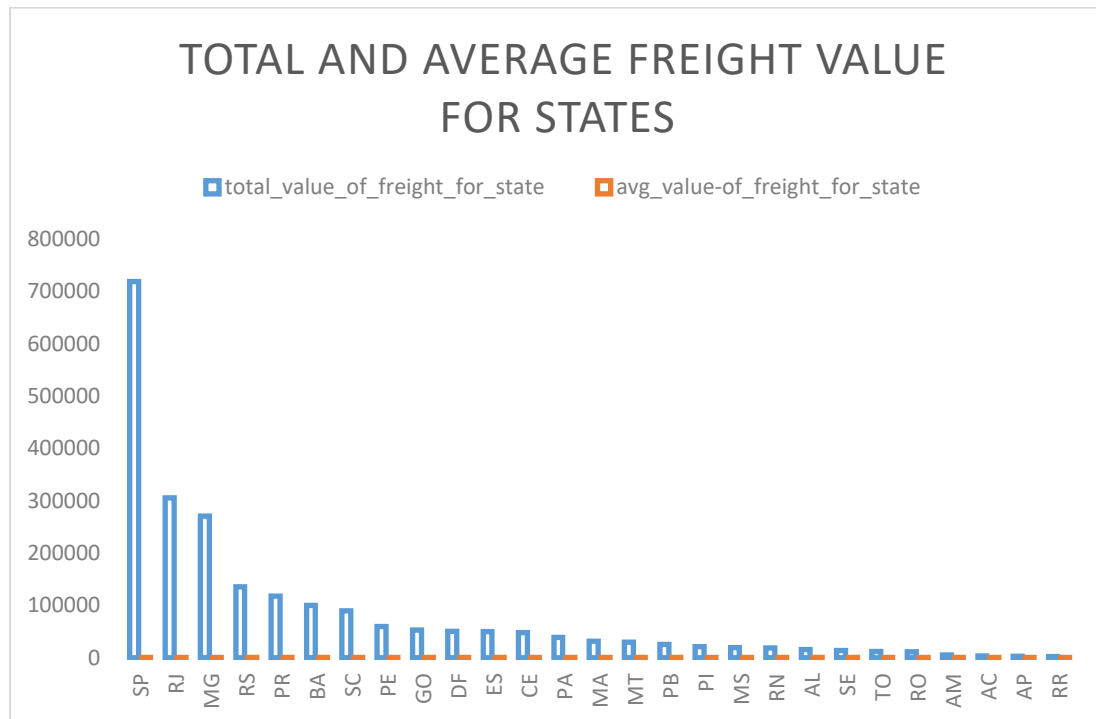
join all the table with specific condition to get state and fright value.

group them on states.

Code-

```
select
c.customer_state,
round(sum(ot.freight_value),2) as `total_value_of_freight_for_state`,
round(avg(ot.freight_value),2) as `avg_value-of_freight_for_state`
from
`target_ecom.order_items` ot inner join `target_ecom.orders` o
on ot.order_id= o.order_id
inner join `target_ecom.customers` c on o.customer_id=c.customer_id
group by
c.customer_state
order by
total_value_of_freight_for_state desc;
```

CHART



TARGET SALES 2017-2018 ANALYSYS

customer_	total_value_of_freight_for_state	avg_value-of_freight_for_state
SP	718723.07	15.15
RJ	305589.31	20.96
MG	270853.46	20.63
RS	135522.74	21.74
PR	117851.68	20.53
BA	100156.68	26.36
SC	89660.26	21.47
PE	59449.66	32.92
GO	53114.98	22.77
DF	50625.5	21.04
ES	49764.6	22.06
CE	48351.59	32.71
PA	38699.3	35.83
MA	31523.77	38.26
MT	29715.43	28.17
PB	25719.73	42.72
PI	21218.2	39.15
MS	19144.03	23.37
RN	18860.1	35.65
AL	15914.59	35.84
SE	14111.47	36.65
TO	11732.68	37.25
RO	11417.38	41.07
AM	5478.89	33.21
AC	3686.75	40.07
AP	2788.5	34.01
RR	2235.19	42.98

FINDINGS –

Maximum freight value is from state SP .

Avg freight value is highest for state RR which have the minimum total freight value.

SP ,RJ, MG ,RS states has the highest total freight value that shows their no of product sold is highest.

#avg freight value is lowest in sp and RJ, MG ,RS that shows the no of orders placed by these state are Highest.

5 Analysis based on sales, freight and delivery time

TARGET SALES 2017-2018 ANALYSIS

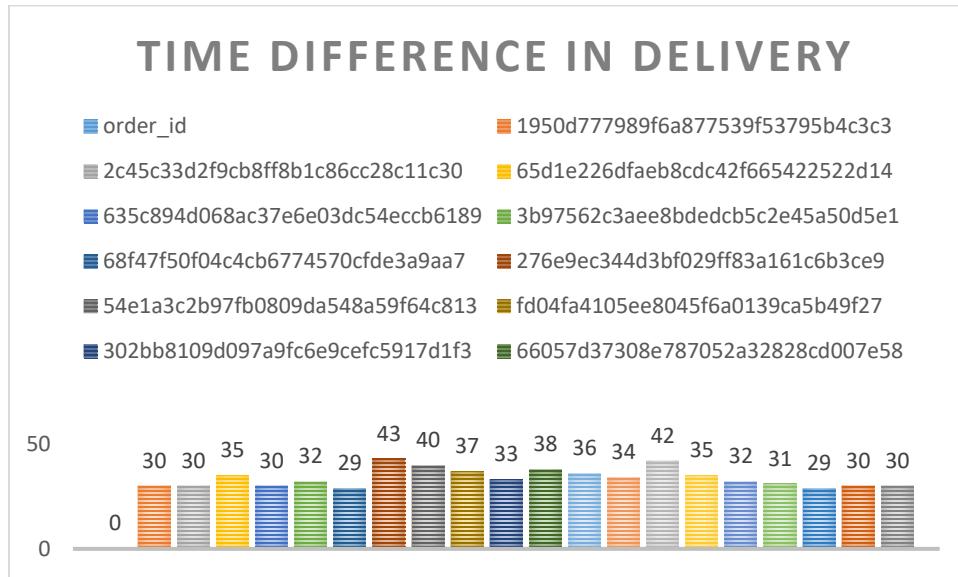
5(1) Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query.

```
Code- SELECT order_id,
date_diff(order_delivered_customer_date, order_purchase_timestamp, day) as `
time_to_deliver`,
date_diff(order_delivered_customer_date, order_estimated_delivery_date , day) as
`diff_estimated_delivery`
from
`target_ecom.orders`
WHERE
order_delivered_customer_date IS NOT NULL
```

TARGET SALES 2017-2018 ANALYSYS

	order_id	time_to_deliver	diff_estimated_delivery
1	1950d777989f6a877539f53795b4c3c3	30	12
2	2c45c33d2f9cb8ff8b1c86cc28c11c30	30	-28
3	65d1e226dfaeb8cdc42f665422522d14	35	-16
4	635c894d068ac37e6e03dc54eccb6189	30	-1
5	3b97562c3aee8bdedcb5c2e45a50d5e1	32	0
6	68f47f50f04c4cb6774570cfde3a9aa7	29	-1
7	276e9ec344d3bf029ff83a161c6b3ce9	43	4
8	54e1a3c2b97fb0809da548a59f64c813	40	4
9	fd04fa4105ee8045f6a0139ca5b49f27	37	1
10	302bb8109d097a9fc6e9cefc5917d1f3	33	5
11	66057d37308e787052a32828cd007e58	38	6
12	19135c945c554eebfd7576c733d5ebdd	36	2
13	4493e45e7ca1084efcd38ddebf174dda	34	0
14	70c77e51e0f179d75a64a614135afb6a	42	11
15	d7918e406132d7c81f1b845276b03a3b	35	3
16	43f6604e77ce6433e7d68dd86db73b45	32	7
17	37073d851c3f30deebe598e5a586bdbd	31	9
18	d064d4d070d914984df25775004fce96	29	0
19	61d430273ff1e88f2944acb53e99eab5	30	0
20	d2f8ef9dd1714fcac7de9f0aef13d21a	30	8
21	81279a15416799e6580df60f66760a7b	31	12
22	c429654419aacfe84ec52dd4c45f064d	36	19
23	3f6da1442aba80bcf61179602dfab9ca	33	6
24	553c468f68af19869694de8bcd095213	34	0
25	c9dbfdbd1f10a0b35896dec49bee8884	31	5
26	d2578728ecc606f42c30e24e6ef79c90	54	22
27	604eadfa781354090d53180d97b11dca	30	2
28	cac7b35f7108f26a0b806132cfee4439	40	12
29			

TARGET SALES 2017-2018 ANALYSYS



5(2) Find out the top 5 states with the highest & lowest average freight value.

Assumptions

first we have to join table which have state column and freight values.

in this given data customer table, order table , order items table will be inner join on common columns.

Find the average value of freight value and group it by state column

Rank the avg freight value on desc order .

To get top 5 limit order to 5.

To get lowest average freight values repeat the process and order by asc order.

union both column

Code –

#Find out the top 5 states with the highest & lowest average freight value.

```
(select
c.customer_state,
round(avg(ot.freight_value),2) as `avg_freight_value`,
dense_rank() over( order by round(avg(ot.freight_value),2) desc) as `ranking`
from `target_ecom.customers` c inner join `target_ecom.orders` o
on c.customer_id=o.customer_id
inner join `target_ecom.order_items` ot on o.order_id=ot.order_id
group by
c.customer_state
order by
```

TARGET SALES 2017-2018 ANALYSYS

```
avg_freight_value desc
limit 5)
union all
( select
c.customer_state,
round(avg(ot.freight_value),2) as `avg_freight_value`,
dense_rank() over( order by round(avg(ot.freight_value),2) asc) as `ranking`
from `target_ecom.customers` c inner join `target_ecom.orders` o
on c.customer_id=o.customer_id
inner join `target_ecom.order_items` ot on o.order_id=ot.order_id
group by
c.customer_state
order by
avg_freight_value asc
limit 5)
table
```

1	customer_state	avg_freight_value	ranking	
2	RR	42.98	1	
3	PB	42.72	2	
4	RO	41.07	3	
5	AC	40.07	4	
6	PI	39.15	5	
7	SP	15.15	1	
8	PR	20.53	2	
9	MG	20.63	3	
10	RJ	20.96	4	
11	DF	21.04	5	
12				
13				

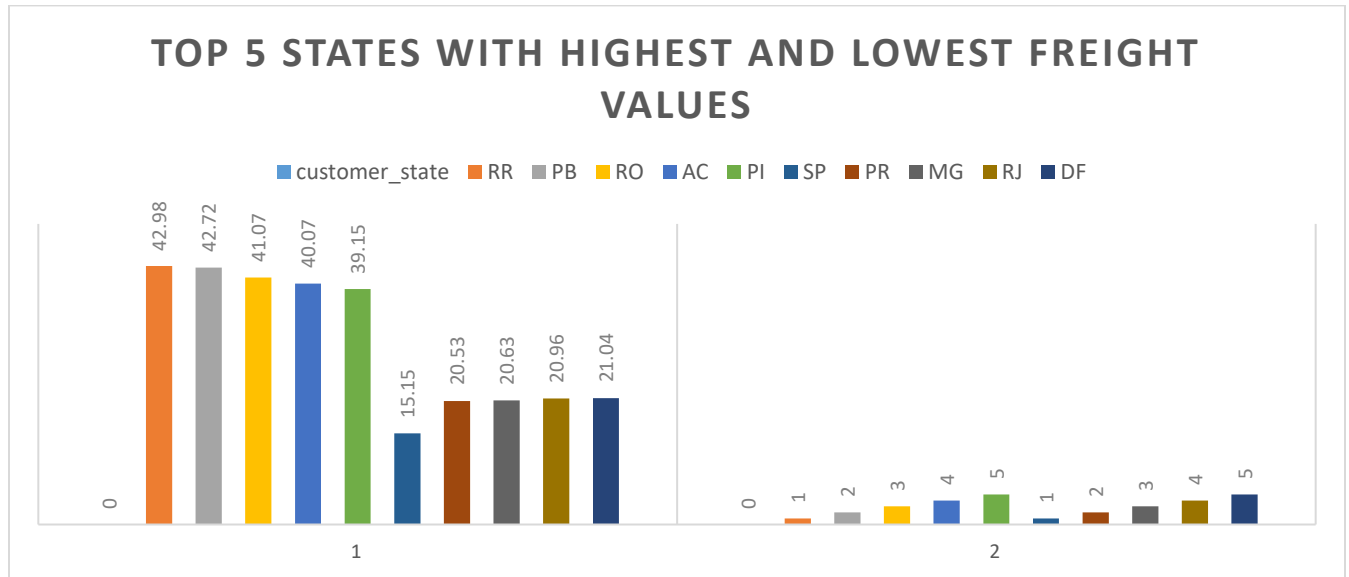
red colour shows top 5 highest avg freight values and their states.

green colour shows top 5 lowest avg freight values and their states.

FINDINGS-

state RR has the highest avg freight value and state SP has the lowest avg freight value.

TARGET SALES 2017-2018 ANALYSYS



chart

5(3) Find out the top 5 states with the highest & lowest average delivery time.

Code-

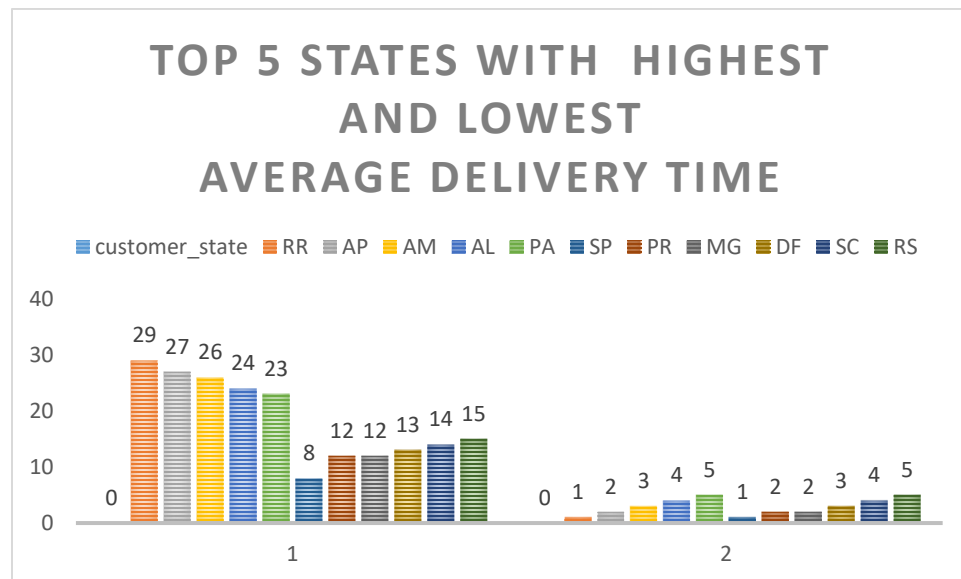
```
#Find out the top 5 states with the highest & lowest average delivery time.
(select
c.customer_state,
round(avg(date_diff(o.order_delivered_customer_date,o.order_purchase_timestamp
,day)),0) as`avg_delivery_time`,
dense_rank()over(order by
round(avg(date_diff(o.order_delivered_customer_date,o.order_purchase_timestamp
,day)),0) desc) as `rank`
from
`target_ecom.orders` o inner join `target_ecom.customers` c
on o.customer_id=c.customer_id
group by
c.customer_state
order by
avg_delivery_time desc
limit 5)
union all
(select
c.customer_state,
round(avg(date_diff(o.order_delivered_customer_date,o.order_purchase_timestamp
,day)),0) as`avg_delivery_time`,
```

TARGET SALES 2017-2018 ANALYSYS

```
dense_rank()over(order by
round(avg(date_diff(o.order_delivered_customer_date,o.order_purchase_timestamp
,day)),0) asc) as `rank`
from
`target_ecom.orders` o inner join `target_ecom.customers` c
on o.customer_id=c.customer_id
group by
c.customer_state
order by
avg_delivery_time asc
limit 6)
```

1	customer_state	avg_delivery_time	rank
2	RR	29	1
3	AP	27	2
4	AM	26	3
5	AL	24	4
6	PA	23	5
7	SP	8	1
8	MG	12	2
9	PR	12	2
10	DF	13	3
11	SC	14	4
12	GO	15	5

output- 13



AVERAGE DELIVERY TIME AS per state

Findings-

TARGET SALES 2017-2018 ANALYSYS

sp has fastest delivery service and RR has slowest delivery service.

State RR has the slowest delivery service

5(4) Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

Assumption

group by states.

find the difference of avg of delivery time on state and of actual delivery time on states.

join table which have column like actual delivery time and state .

code –

```
WITH `cte` as(select c.customer_state,
round(avg(date_diff(order_estimated_delivery_date,order_delivered_customer_date,day)),
2)as `avg_delivery_date`,
dense_rank()over(order by
round(avg(date_diff(order_estimated_delivery_date,order_delivered_customer_date,day)),
2) desc ) as `rank`
from `target_ecom.orders` o inner join `target_ecom.customers` c
on o.customer_id=c.customer_id
where
o.order_status = "delivered" AND
order_delivered_customer_date IS NOT NULL
group by
c.customer_state
order by
avg_delivery_date desc)

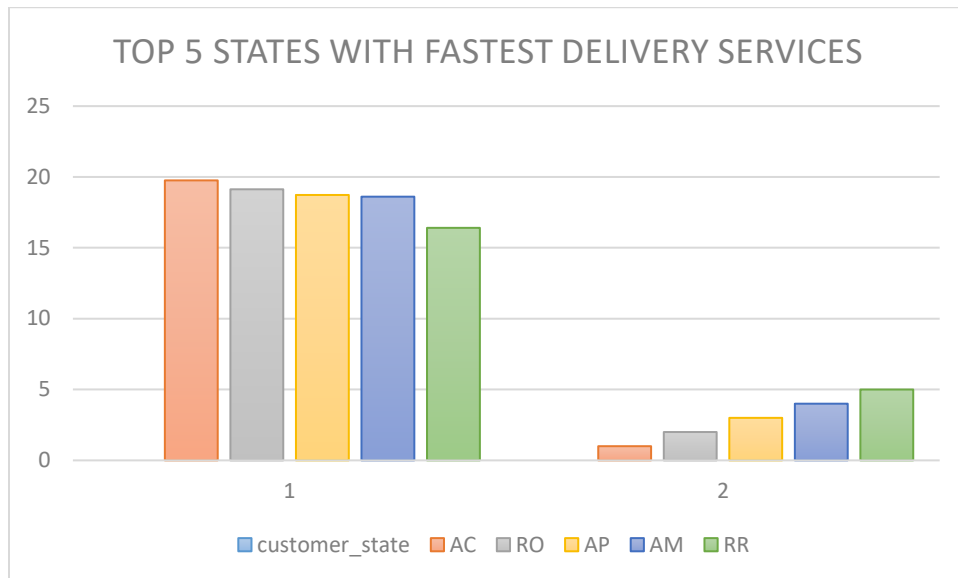
select
* from cte
where
rank<=5
```

.

output

1	customer_state	avg_delivery_date	rank
2	AC	19.76	1
3	RO	19.13	2
4	AP	18.73	3
5	AM	18.61	4
6	RR	16.41	5
7			

TARGET SALES 2017-2018 ANALYSYS



FINDINGS

STATE AC has the fastest avg delivery service as compared to other state.

Top 5 states with fastest delivery services are AC ,RO,AP,AM,RR.

6 Analysis based on the payments:

6(1) Find the month on month no. of orders placed using different payment types.

CODE –

```
select
extract(month from o.order_purchase_timestamp) as `month`,
p.payment_type,
count(p.payment_type)as `no_of_orders_placed`

from `target_ecom.orders` o inner join `target_ecom.payments` p
on o.order_id=p.order_id
group by
extract(month from o.order_purchase_timestamp),p.payment_type
order by
no_of_orders_placed desc;
```


TARGET SALES 2017-2018 ANALYSYS

output

1	month	payment_type	no_of_orders_placer
2		5 credit_card	8350
3		8 credit_card	8269
4		7 credit_card	7841
5		3 credit_card	7707
6		4 credit_card	7301
7		6 credit_card	7276
8		2 credit_card	6609
9		1 credit_card	6103
10		11 credit_card	5897
11		12 credit_card	4378
12		10 credit_card	3778
13		9 credit_card	3286
14		8 UPI	2077
15		7 UPI	2074
16		5 UPI	2035
17		3 UPI	1942
18		6 UPI	1807
19		4 UPI	1783
20		2 UPI	1723
21		1 UPI	1715
22		11 UPI	1509
23		12 UPI	1160
24		10 UPI	1056

CHART

[illegible]

MAXIMUM orders placed by customers by credit card.

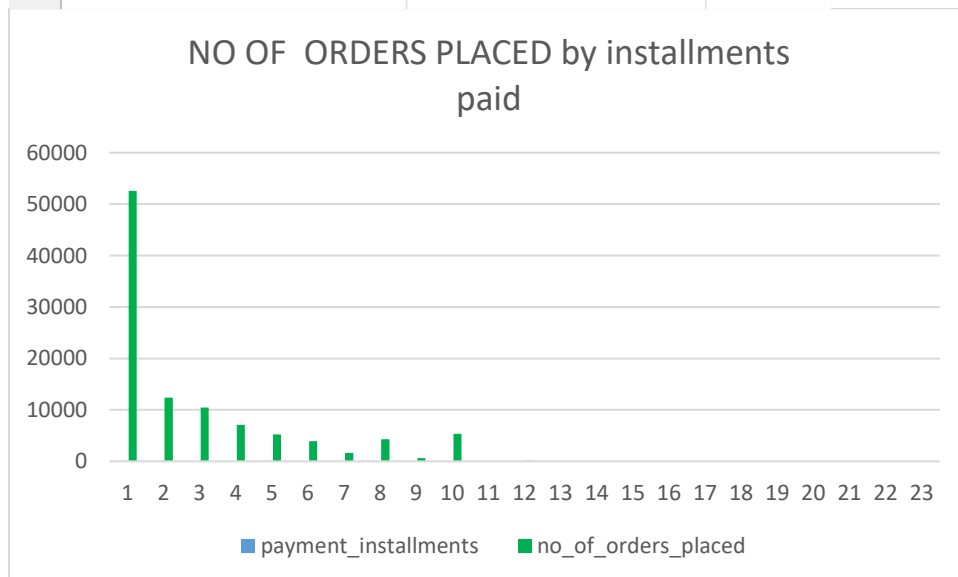
maximum order placed by credit card from January to August month

Code-

```
select
payment_installments,
count(order_id) as `no_of_orders_placed`
from `target_ecom.payments`
group by
payment_installments
having
payment_installments > 0
```

TARGET SALES 2017-2018 ANALYSYS

1	payment_installments	no_of_orders_placed
2	1	52546
3	2	12413
4	3	10461
5	4	7098
6	5	5239
7	6	3920
8	7	1626
9	8	4268
10	9	644
11	10	5328
12	11	23
13	12	133
14	13	16
15	14	15
16	15	74
17	16	5
18	17	8
19	18	27
20	20	17
21	21	3
22	22	1
23	23	1
24	24	18



RECOMMENDATIONS

1. No of orders placed are maximum in months march, April, May, July, August and November rest months need to take care to increase sale.
2. Minimum orders are placed in September and October months of the year, so the target need to focus on these months and make some offers to increase sale.
3. During DAWN time of the day minimum orders are placed so target need to make some special offers so customer will attract to place orders in dawn time of the day also.
4. There are some states where month on month orders placed always minimum because no of customers are less from those state .states are TO,RO,RR,PB,PA,MT,MS,GO here no of customers very less to attracts customers from these specific states company need to make some decision.
5. Across the states customers are maximum from SP,RJ,MG,RS PR
6. Minimum customers are from RN,SE,TO,RO,AM,AC,AP these states need special attention to increase sales for upcoming years.
7. Sales increase for year 2017 to 2018 for specific January to August is 136%.
8. TOTAL value of order price is maximum for state SP, THAT indicates Maximum customers are from state SP.
9. Average value of order price is maximum for PB ,AL states.
10. MONTH ON MONTH ORDERS PLACED where maximum payment paid by customers by credit card, then upi is mode of payment.
11. 1 to 8 installment paid by maximum number of customers for further company need to give attention.

TARGET SALES 2017-2018 ANALYSYS